HDB Resale Price Statistical Report - Nabil Rakaiza Abror | A0305715J
ST3131 : Regression Analysis - Assignment

A. Introduction

a. Introduction

The resale price of Housing & Development Board (HDB) flats in Singapore has been a subject of extensive research over the years. Such analysis is valuable for both prospective sellers seeking to determine an appropriate valuation and buyers aiming to assess fair market prices. The primary objective of this report is to develop a linear regression model that provides accurate price predictions while maintaining interpretability, minimizing multicollinearity, and adhering to all fundamental statistical assumptions. Utilizing a dataset of HDB resale transactions from July 2020, the study starts with initial data preprocessing and is followed by exploratory data analysis (EDA). Finally, the model is rigorously evaluated to propose an optimized final version based on the preceding diagnostic findings.

B. Data Preprocessing and Exploratory Data Analysis

a. Data Preprocessing

Since the dataset is not inherently structured for linear regression, several preprocessing steps were implemented. These transformations ensure the data conforms to the requirements of a linear model and improve the overall predictive power.

1. Exclusion of Constant Features
   The month column was removed from the dataset. Since the data is restricted exclusively to July 2020, this feature contains only a single unique value, providing no predictive power.
2. Standardization of Lease Temporal Data
   The remaining_lease attribute was originally formatted as a string. To preserve its numerical significance, Regular Expressions (regex) were employed to extract and convert these values into a continuous numerical format representing the total number of months.
3. Engineering of Spatial and Geographic Features
   The raw block and street_name columns are high-cardinality categorical variables that are difficult to interpret within a linear framework. To derive more meaningful predictors, the OneMap API was utilized to retrieve the precise latitude, longitude, and postal codes for each entry. These coordinates enabled the calculation of proximity-based features, such as distance to the central business district (dist_to_cbd) and distance to the nearest MRT station (dist_to_mrt).
4. Transformation of Storey Ranges
   The storey_range variable was initially provided in binned intervals. Treating these ranges as purely categorical would ignore the inherent ordinal and numerical nature of floor levels. To resolve this, the midpoint value of each range was extracted to create a discrete numerical variable. This conversion allows the model to more efficiently quantify the height of HDB as the regressors.

Declaration: I declare that I used an AI tool (Gemini) to assist with fixing coding errors, generating portions of the code, validating my analytical ideas, and reviewing my report for grammatical and stylistic improvements. My original work contributes to at least 50% of this report.

b. Exploratory Data Analysis

Exploratory Data Analysis (EDA) provides a foundational understanding of variable distributions and latent inter-relationships. Various visualization techniques were employed to characterize the data and detect early indicators of multicollinearity.

1. Univariate Analysis: Histogram Findings

   The histograms presented in Figure 1 offer critical insights into the distributional characteristics of the target variable and its predictors. The target variable, resale_price, exhibits a distinct positive skew, while floor_area_sqm and lat appear to approximate a normal distribution. Similarly, dist_to_mrt and storey_avg demonstrate right-skewed patterns. In contrast, lon follows a bimodal distribution, suggesting the presence of two primary clusters. For remaining_lease and lease_commence_date, however, the distributional forms remain indeterminate and do not clearly align with standard parametric archetypes.

2. Bivariate Analysis: Numerical Predictors vs. Target

   Scatterplots in Figure 2 were employed to evaluate the strength and direction of the relationships between the independent variables and resale_price. A discernible positive association exists between resale_price and both floor_area_sqm and storey_avg, suggesting that larger areas and higher floor levels correspond with increased valuations. Conversely, dist_to_mrt and dist_to_cbd exhibit negative correlations with price, suggesting that places nearer to MRT/CBD have relatively higher prices. While most predictors demonstrate low-to-moderate linear relationships with the target variable, significant heteroscedasticity is evident. The observed dispersion indicates that the variance in resale_price fluctuates across different levels of the predictors.

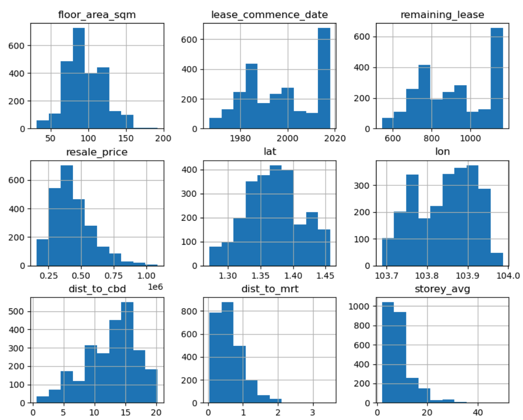

Figure 1. Histogram

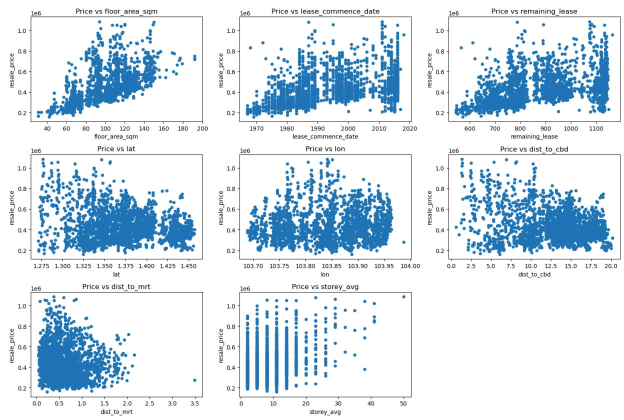

Figure 2. Scatter Plot of Numerical vs Target

3. Categorical Analysis: Boxplot Findings

   Boxplots in Figure 3 were utilized to evaluate the influence of categorical attributes on property valuations. A consistent positive trend is observed regarding storey_range and room_type, where units on higher floors and those with greater room capacities command significantly higher prices. Spatially, resale_price exhibits substantial variation across different towns, corroborating that geographical location serves as a primary determinant of market value. Furthermore, notable price variance exists across flat_model types, with specialized models consistently demonstrating higher valuations relative to standard apartment configurations.

4. Multicollinearity
   In light of the multicollinearity between lease_commence_date and remaining_lease in Figure 4, one of these features was excluded prior to model estimation. Removing redundant predictors is essential to prevent inflated standard errors and ensure that the individual coefficients of the final linear model remain interpretable and statistically sound.

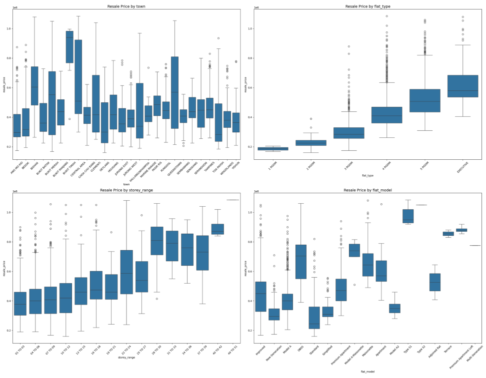

Figure 3. Boxplot

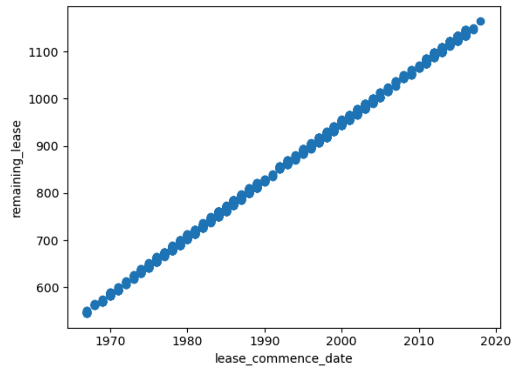

Figure 4. Detected Multicollinearity

C. Modelling

a. Baseline Model Specification

The initial phase involves fitting an Ordinary Least Squares (OLS) Multiple Linear Regression (MLR) model to establish a performance baseline. The features selected for this model include town, flat_type, flat_model, storey_avg, floor_area_sqm, remaining_lease, dist_to_cbd, dist_to_mrt, lat, and lon. These variables were selected because they represent the core structural and locational determinants of property value. Features such as postal_code were excluded due to their lack of numerical significance, while other high-cardinality categorical columns were removed to prevent overfitting and maintain model parsimony.

1. Goodness of Fit and ANOVA
   The baseline model achieved an R^2 of 0.897 and an adjusted R^2 of 0.895. The ANOVA results indicate that the overall regression is statistically significant ($p < 0.001$), with the exception of lat and lon coefficients that were not statistically significant. This suggests that the coordinate data is likely redundant once more specific spatial proxies are included. Consequently, these geographic coordinates will be removed in subsequent iterations.

2. Multicollinearity Assessment (VIF)
   A Variance Inflation Factor (VIF) analysis was conducted to identify potential dependencies between independent variables. Several features, specifically dist_to_cbd, town, flat_type, flat_model, and floor_area_sqm, exhibited elevated VIF values. The rationale for this multicollinearity is grounded in the

nature of Singapore's urban planning. dist_to_cbd is inherently tied to the town variable, as a specific town's location largely dictates its distance from the city center. In addition, flat_type and floor_area_sqm are naturally correlated, for instance, a 5-room flat will almost always have a larger square footage than a 3-room flat. Similarly, certain flat_model designations are systematically associated with larger floor areas compared to standard models.

3. Residual diagnostics
   Diagnostic plots in Figure 5 indicate that the baseline model violates several key assumptions. The Q-Q plot reveals a "heavy tail" on the right side, where residuals deviate upward from the theoretical line, suggesting the model violates the normality assumptions. The residuals-versus-fitted plot displays a hybrid "U-shape" and "funnel" pattern, indicating non-constant variance (heteroscedasticity) and suggests that the relationship between the predictors and the target may not be strictly linear, or that a transformation of the target variable is required to stabilize the variance.

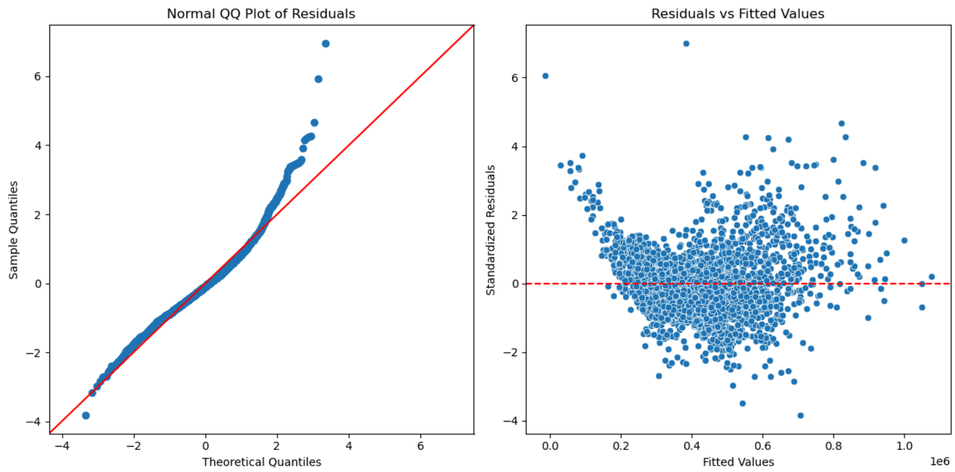

Figure 5. Initial model QQ Plot and Scatter Plot of Residual vs Fitted Values

b. Addressing Model Violations

The initial model diagnostics revealed significant violations of the assumptions, specifically multicollinearity, heteroscedasticity, and non-normality. The following sections detail the corrective measures taken to ensure the model's statistical validity.

1. Mitigating Multicollinearity
   To address high Variance Inflation Factors (VIF), two distinct strategies were explored: Model Respecification and Ridge Regression.
   - Model Respecification
     An iterative elimination process was conducted to identify redundant features within two primary collinear groups:
     - Structural Group: (flat_type, flat_model, floor_area_sqm)
     - Locational Group: (dist_to_cbd, town)

     Through empirical testing, floor_area_sqm and town were retained as the primary representatives for their respective groups. This

configuration was selected as it optimized the balance between explanatory power (R^2) and model parsimony, effectively reducing VIF values to acceptable levels without sacrificing significant predictive accuracy.

- Ridge Regression
  As an alternative, a Ridge Regression model was fitted to the unpruned feature space, achieving a higher R^2 of 92.47%. By shrinking rather than eliminating correlated predictors, Ridge provides a more robust and generalizable framework. However, this comes at the cost of interpretability. The introduction of the L2 penalty introduces bias into the coefficients, making the exact marginal financial impact of any single predictor less transparent than in a standard OLS model. Given that our primary objective requires clear, actionable insights into specific financial drivers, the Ridge approach was not selected for the final model.

2. Resolving Heteroscedasticity and Normality
   To correct the "U-shape" and "funnel" patterns observed in the residual plots, a logarithmic transformation was applied to the target variable (resale_price). The transformation successfully stabilized the variance. The updated residuals-versus-fitted plot demonstrates a random dispersion of data points, indicating that the assumption of homoscedasticity has been largely satisfied. The Normal Q-Q plot still exhibits a slight upward deviation in the right tail. This indicates that while the central distribution of errors is nearly normal, outliers persist in the upper quantiles.

   While further outlier removal would yield a cleaner Q-Q plot and improve normality metrics, these extreme values represent real-world market volatility. Removing them could result in a model that offers a false sense of precision and fails to account for the high-end premiums typical of the Singapore property market. Consequently, these observations were retained to maintain the model's ecological validity.

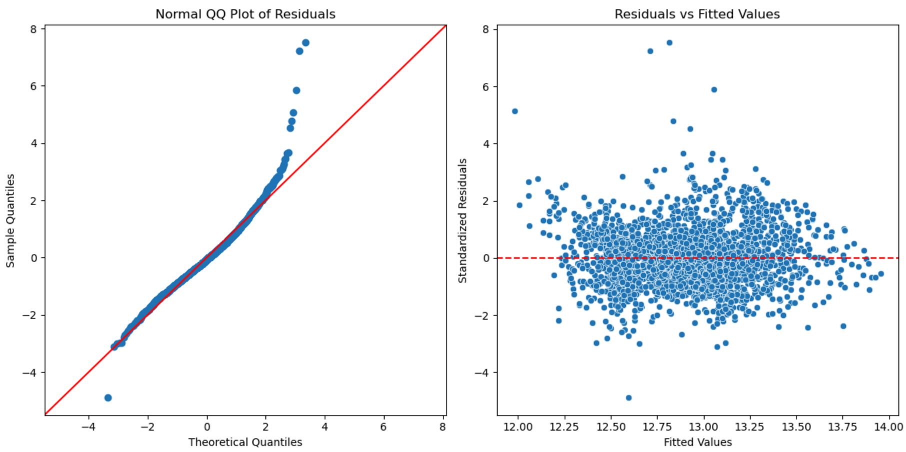

Figure 6. Intermediate model QQ Plot and Scatter Plot of Residual vs Fitted Values

c. Proposed Model

Based on the diagnostic findings, an Ordinary Least Squares (OLS) framework utilizing a log-transformed target variable is proposed as the final model. The feature set was optimized to include town, remaining_lease, dist_to_mrt, floor_area_sqm, and storey_avg. The application of a logarithmic transformation to the resale price was instrumental in mitigating the heteroscedasticity and non-linearity observed in the baseline diagnostics, thereby ensuring the model adheres to the fundamental assumptions required for reliable statistical inference.

As an example on how this model is interpreted, when holding all other factors constant, a one-month increase in remaining_lease is associated with an approximate 0.11% increase in the resale price. Similarly, when controlling for all other variables, a property located in Clementi is expected to be approximately 15.05% less expensive than a comparable unit in Bukit Timah. The comprehensive ANOVA results and model coefficients are presented in Figure 7 and Figure 8, respectively.

OLS Regression Results			
Dep. Variable:	np.log(resale_price)	R-squared:	0.896
Model:	OLS	Adj. R-squared:	0.895
Method:	Least Squares	F-statistic:	720.1
Date:	Wed, 08 Apr 2026	Prob (F-statistic):	0.00
Time:	18:11:46	Log-Likelihood:	1975.6
No. Observations:	2456	AIC:	-3891.
Df Residuals:	2426	BIC:	-3717.
Df Model:	29		
Covariance Type:	nonrobust		

Figure 7. ANOVA table of final model

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	11.1053	0.019	592.331	0.000	11.069	11.142
C(town)[T.BEDOK]	-0.0140	0.014	-0.984	0.325	-0.042	0.014
C(town)[T.BISHAN]	0.2189	0.024	9.302	0.000	0.173	0.265
C(town)[T.BUKIT BATOK]	-0.1790	0.017	-10.775	0.000	-0.212	-0.146
C(town)[T.BUKIT MERAH]	0.1958	0.017	11.445	0.000	0.162	0.229
C(town)[T.BUKIT PANJANG]	-0.3408	0.016	-21.878	0.000	-0.371	-0.310
C(town)[T.BUKIT TIMAH]	0.2789	0.046	6.050	0.000	0.189	0.369
C(town)[T.CENTRAL AREA]	0.3349	0.028	12.148	0.000	0.281	0.389
C(town)[T.CHOA CHU KANG]	-0.3792	0.016	-23.945	0.000	-0.410	-0.348
C(town)[T.CLEMENTI]	0.1157	0.017	6.827	0.000	0.082	0.149
C(town)[T.GEYLANG]	-0.0022	0.018	-0.121	0.903	-0.037	0.033
C(town)[T.HOUGANG]	-0.1028	0.015	-6.821	0.000	-0.132	-0.073
C(town)[T.JURONG EAST]	-0.1316	0.019	-6.976	0.000	-0.169	-0.095
C(town)[T.JURONG WEST]	-0.2165	0.015	-14.702	0.000	-0.245	-0.188
C(town)[T.KALLANG/WHAMPOA]	0.0981	0.020	4.819	0.000	0.058	0.138
C(town)[T.MARINE PARADE]	0.3242	0.033	9.693	0.000	0.259	0.390
C(town)[T.PASIR RIS]	-0.1533	0.016	-9.307	0.000	-0.186	-0.121
C(town)[T.PUNGGOL]	-0.2881	0.015	-19.116	0.000	-0.318	-0.259
C(town)[T.QUEENSTOWN]	0.2136	0.018	11.956	0.000	0.179	0.249
C(town)[T.SEMBAWANG]	-0.4086	0.018	-22.255	0.000	-0.445	-0.373
C(town)[T.SENGKANG]	-0.3472	0.015	-23.159	0.000	-0.377	-0.318
C(town)[T.SERANGOON]	0.0836	0.022	3.775	0.000	0.040	0.127
C(town)[T.TAMPINES]	-0.0661	0.014	-4.641	0.000	-0.094	-0.038
C(town)[T.TOA PAYOH]	0.0429	0.017	2.554	0.011	0.010	0.076
C(town)[T.WOODLANDS]	-0.3548	0.015	-24.404	0.000	-0.383	-0.326
C(town)[T.YISHUN]	-0.2137	0.014	-14.805	0.000	-0.242	-0.185
storey_avg	0.0085	0.000	19.871	0.000	0.008	0.009
floor_area_sqm	0.0106	9.6e-05	110.042	0.000	0.010	0.011
remaining_lease	0.0011	1.75e-05	61.073	0.000	0.001	0.001
dist_to_mrt	-0.1370	0.007	-18.839	0.000	-0.151	-0.123

Figure 8. Coefficients of final model

D. Conclusion

The log-transformed OLS model is selected as the optimal model, explaining 89.5% of the variance in HDB resale prices through five predictors: town, floor area, remaining lease, average storey, and distance to MRT. This model successfully addresses initial issues of multicollinearity and heteroscedasticity. While minor non-normality persists in the upper residual tail, these deviations are considered acceptable as they represent genuine transactions of ultra-premium flats rather than data errors. Future research could enhance the model's predictive power by incorporating granular proximity data to amenities like primary schools and shopping malls. Additionally, integrating macroeconomic factors such as interest rate fluctuations would allow the model to adapt to broader shifts in Singapore's property market.